and take snapshots using the snapshot lifecycle management (SLM) feature. Elasticsearch provides the `_slm` API to manage the lifecycle of snapshots and create lifecycle policies that are executed on a predefined schedule.

To do this, we can create a policy using the `_slm` API. This policy contains information such as the indexes it should back up, the schedule (cron job), the retention period, and so on. We must have registered a repository (see section 14.3.2), because it is a prerequisite for snapshot lifecycle management.

Suppose we want to back up all our movie indexes at midnight every night to a given repository. We also need to keep these snapshots for a week. We can write a policy and create the automation using the `_slm` API.

Listing 14.11 Creating a policy for scheduled snapshots

```
PUT _slm/policy/prod_cluster_daily_backups
{
  "name":"<prod_daily_backups-{now/d}>",
  "schedule": "0 0 0 * * ?",
  "repository": "es_cluster_snapshot_repository",
  "config": {
    "indices":["*movies*", "*reviews*"],
    "include_global_state": false
  },
  "retention":{
    "Expire_after":"7d"
  }
}
```

The _slm API expects a policy identifier.

Unique name given to a snapshot

Registered repository

Cron job schedule for midnight

Indexes to snapshot

Adds a cluster state to the snapshot

Retains the snapshot for a week (7d)

The `_slm` API creates a policy that is stored in the cluster for execution when the schedule kicks in. We must provide three parts: a unique name, the schedule, and the repository we registered earlier to store the snapshots. Let's look at these bits in detail.

The unique name (`<prod_daily_backups-{now/d}>` in listing 14.11) is a name constructed with data math in it. In this case, `<prod_daily_backups-{now/d}>` is parsed to `prod_daily_backups-5.10.2022` if it is run on October 5, 2022, because `{now/d}` indicates the current date. Every time the schedule kicks in, a new unique name is generated with the current date: `prod_daily_backups-6.10.2022`, `prod_daily_backups-7.10.2022`, and so on. Because we use date math in the name, we must enclose the name using angle brackets (`< >`) so the parser parses without any problems. Consult the Elasticsearch documentation (http://mng.bz/QPwe) for further details on date math in names.

As listing 14.11 shows, we provided a schedule in the form of a cron job: `"schedule": 0 0 0 * * ?`. This cron expression states that the job should be executed precisely at midnight. We can therefore expect our snapshot process to begin at midnight every night.

The `config` block in listing 14.11 consists of the indexes and cluster state we want to back up (in this example, all movies and review-related indexes). If we do not